9.10. ABSOLUTE DISINTEGRATION RATES

The determination of absolute disintegration rates is of great importance in all areas of nuclear chemistry, tracer work, age calculation, etc. Numerous methods have been employed, many using techniques described above, as track counting, liquid scintillation measurements, 4π proportional counters, etc. If the nuclei decay through β-γ emission, the absolute rate may be obtained by two detectors placed close to each side of a thin sample, one detector β-sensitive and the other γ-sensitive.

When only a single detector in a conventional counting set-up is available (e.g. detector arrangement in Fig. 9.4), absolute counting rates can be obtained for unknown samples by comparison with known standards.

When standards are not available it is possible to obtain an approximate estimation of the absolute disintegration rate from a knowledge of the various factors that influence the counting efficiency. The detection efficiency ψ is defined as a ratio between the count rate and the absolute disintegration rate Eqn (5.64). This detection efficiency, which was discussed briefly in section 7.2, is the product of all the factors which influence the measured count rate and may be expressed as

$$\psi = \psi_{\text{det}}\psi_{\text{res}}\psi_{\text{geom}}\psi_{\text{back}}\psi_{\text{self}}\psi_{\text{abs}} \tag{9.32}$$

where

ψ_{det} = counting efficiency of detector,
ψ_{res} = resolving time correction (see section 9.3),
ψ_{geom} = geometry factor (see section 7.2),
ψ_{back} = backscattering factor,
ψ_{self} = self-absorption factor ($\psi_{\text{sample}} = \psi_{\text{back}}\psi_{\text{self}}$), and
ψ_{abs} = absorption factor (see section 7.2).

The efficiency of the detector is a measure of the number of counts registered compared to the number of particles that enter the sensitive volume of the detector. This efficiency is approximately 100% for α- and high-energy β-particles in most detectors, but often substantially lower for γ-rays. In as much as it is quite difficult to apply simple geometric considerations to the solid angle subtended by a detector for a source which is not concentrated at a point the factors ψ_{det} and ψ_{geom} may be determined experimentally by using a very thin standard source of approximately the same area as the unknown. The factor ψ_{geom} can be calculated for circular samples and detector windows, see section 7.2

It was noted in section 7.4 that β-rays undergo large angle deflections. As a result, β-particles from the sample which may start in a direction away from detector can be deflected by several scattering events back into the detector. Such backscattering is dependent upon the atomic number of the material upon which the sample is supported (cf. Fig. 7.14). ψ_{back} increases with backing material thickness up to a saturation thickness beyond which it is constant. Counting is usually done with either an

essentially weightless backing ($\psi_{back} = 1$) or with a backing sufficiently thick as to have saturation. The saturation thickness corresponds to approximately 20% of the range of the β-particles in the backing material. Scattering can also occur from the walls of the sample holder, but this is usually less important than the backscattering from the sample backing material.

If the sample is thick, the count rate may be increased by internal backscattering, but the decrease of the count rate due to self-absorption within the sample is a greater factor. This sample self-absorption is inversely proportional to the β-ray energy and directly proportional to the thickness of the sample. For β^- from ^{32}P (1.72 MeV) sample thicknesses of 15 mg cm^{-2} show little self-absorption, while for β^- from ^{14}C (0.15 MeV), the absorption for sample thicknesses as small as 1 mg cm^{-2} is significant. The absorption factor can be determined by counting a series of samples of different thicknesses but with the same total disintegration rate, and then extrapolating that to zero sample weight.

The absorption factor ψ_{abs} is related to the absorption of the particles after they leave the sample by any covering over the sample, by the air between the sample and the detector, by the absorption in the detector window, etc. Again, this factor is usually determined by experimental comparison with a sample of known absolute disintegration rate, a *standard*.

9.11. SAMPLE PREPARATION

From the discussion of the factors that enter into the counting efficiency it is obvious that the preparation of the counting sample must be done with care and must be reproducible if several samples are to be compared. Counting of α- and β-emitters *in solution* is best achieved by means of liquid scintillation counting. Because in this technique the emitters are included in the detection system itself the efficiency is very high and reproducible.

The advantage of using *solid samples* for counting is that the samples can be made very robust and small, allowing the use of either very simple counting systems (e.g. GM-counters), or the use of commercial very efficient high-capacity (>1000 samples/hour) automatic counting systems. Alpha-emitters can only be counted efficiently if the sample is very thin so that the self-absorbtion is eliminated. For α-spectrometry the surface density should be <0.1 mg cm^{-2}. Preferably, α's should be counted by surface barrier detectors, windowless proportional counters or internal ion chambers. Counting of solid samples of β-emitters may or may not be a problem depending on the energy of the β-emission. Again, care must be taken with uniform thickness of sample, back-scattering, etc. The use of energy sensitive detectors makes possible a reliable measurement of one particular radioactive nuclide in the presence of other radiation of secondary importance.